Who Voted for Whom in 2024? Demographics, Social Class and Party Identification in U.S. Presidential Vote Choice

STAT 107 Half-Way Project Report

November 6, 2026

Abstract

This project uses data from the 2024 American National Election Studies (ANES) to ask which demographic characteristics and political identities are associated with voting for the Democratic or the Republican candidate in the 2024 U.S. presidential election. After cleaning the survey data, we kept the 3,603 respondents who reported voting for one of the two major-party candidates and studied their age, gender, race and ethnicity, education, self-reported social class and party identification. Preliminary visualizations and tests show that vote choice differs significantly across every characteristic we examined, although the age difference is small. Party identification has by far the strongest relationship with vote choice, since about 98% of Democratic identifiers and 94% of Republican identifiers voted for their party's candidate. Among the demographic characteristics, race, education and gender show the largest differences, while age has a small effect and social class follows a non-linear pattern. These results are associations in an unweighted sample and do not show causation. Next, we plan to fit logistic regression models with and without party identification, evaluate their predictive accuracy with cross-validation, and check the robustness of the results with survey weights.

1 Introduction

Every presidential election raises the same question: who votes for which candidate, and why? Differences in vote choice across groups of voters are central to how political scientists, journalists and campaigns understand an election. Our project studies this question for the 2024 U.S. presidential election. We ask which characteristics of a voter (age, gender, race and ethnicity, education, self-reported social class and party identification) are associated with voting for the Democratic candidate rather than the Republican candidate, and how well these characteristics together can predict a voter's choice.

The analysis has two goals. The first is to test hypotheses about individual characteristics, for example whether vote choice differs by education or whether older voters differ from younger voters. The second is to build a classification model that predicts the two-party vote from voter characteristics and to measure how accurate it is. The results could help students and researchers of elections understand which divisions between voters are large and which are small, and they show how much of vote choice can be explained by demographics alone. An important part of the project is to compare models with and without party identification, because party identification is so closely tied to vote choice that it can hide the role of the other characteristics.

This half-way report describes the data and the cleaning steps, presents preliminary visualizations and tests for each characteristic, and lays out the models and further visualizations we plan to complete for the final report.

2 Executive Summary

- Of the 3,603 respondents who voted for a Democratic or Republican presidential candidate, 55.9% voted Democratic and 44.1% voted Republican, so the two outcome groups are reasonably balanced.
- Party identification is almost a direct copy of the vote: 97.7% of Democratic identifiers voted Democratic and 93.5% of Republican identifiers voted Republican. Because of this, models will be fit both with and without it.
- Among the demographic characteristics, race, education and gender show the largest differences in vote choice (for example, 92.3% of Black respondents voted Democratic, 65.8% of respondents with a bachelor's degree or higher voted Democratic compared with 39.7% to 47.9% in the other education groups, and 60.7% of women voted Democratic compared with 49.8% of men). All three differences are statistically significant ($p < .001$).
- Social class does not follow a steady trend: working-class respondents were the only group with a Republican majority (51.3%), and upper-class respondents were the most Democratic (76.2%). The age difference is small: Republican voters were only about 1.3 years older on average, although respondents aged 18 to 29 were more Democratic (64.1%) than respondents aged 60 and older (54.6%).
- Next, we will fit logistic regression models, evaluate them with cross-validation and check the results using survey weights and alternative handling of missing data.

3 Data

3.1 Source and variables

The data come from the American National Election Studies (ANES) 2024 Time Series Study (https://electionstudies.org/), a survey of U.S. adults conducted before and after the 2024 election. The survey was not generated by a randomized experiment or a simulation, so all of our results describe associations among survey responses. The raw file `anes.csv` contains a very large number of survey questions. The file `11_DataCleaning.Rmd` keeps ten of them and renames them, and `21_DataProcessing.Rmd` sets the order of the categories and adds an age group variable. The final analysis data set `anes_processed` has 3,603 observations (respondents) and 9 variables, described below.

Variable	Role	Description	Values after cleaning
`vote_pres`	Outcome	Presidential vote in 2024	Democrat, Republican
`vote_house`	Secondary outcome	U.S. House vote in 2024	Democrat, Republican, Other
`age`	Predictor	Age in years	18 to 80 (ages 80 and older appear to be recorded as 80)
`age_group`	Predictor	Age group, created from `age`	18-29, 30-44, 45-59, 60+
`gender`	Predictor	Gender	Man, Woman, Nonbinary, Other
`race`	Predictor	Race and ethnicity	White, Black, Hispanic, Asian, Middle Eastern/North African, Multiracial/Other

Variable	Role	Description	Values after cleaning
`education_level`	Predictor	Highest level of education	Less than HS, High School/GED, College/Associate, Bachelors or Higher
`social_class`	Predictor	Self-reported social class	Lower Class, Working Class, Middle Class, Upper Class
`party_id`	Predictor	Party identification	Democrat, Republican, Independent/None, Other

3.2 Cleaning steps

The cleaning steps in `11_DataCleaning.Rmd` were the following.

1. Select the ten survey variables of interest and give them readable names.
2. Keep only respondents who reported voting for the Democratic or the Republican candidate for president. This removes non-voters, voters for other candidates and respondents with missing responses, and it makes the outcome a two-category variable.
3. Recode the numeric survey codes into labeled categories for the House vote, gender, social class, education and party identification. For the House vote, gender and party identification, any code that does not belong to a named category is grouped as "Other".
4. Combine the ANES race summary variable and the Middle Eastern/North African indicator into one race and ethnicity variable with six categories (respondents who identified as Middle Eastern/North African are placed in that category), and set negative age codes (refusals) to missing.
5. Convert the education and race variables to factors with a fixed order of levels, and save the cleaned data as `12_cleanedElectionData.RData`.
6. In 21_DataProcessing.Rmd, set a fixed order of categories for social class, gender and party identification, create the age group variable, and save the processed data as 22_processed-ElectionData.RData.

3.3 Missing values

Only a few variables have missing values after cleaning (shown below).

Variable	Missing
vote_pres	0
vote_house	0
gender	0
education_level	50
age	156
party_id	0
social_class	35
race	29
age_group	156

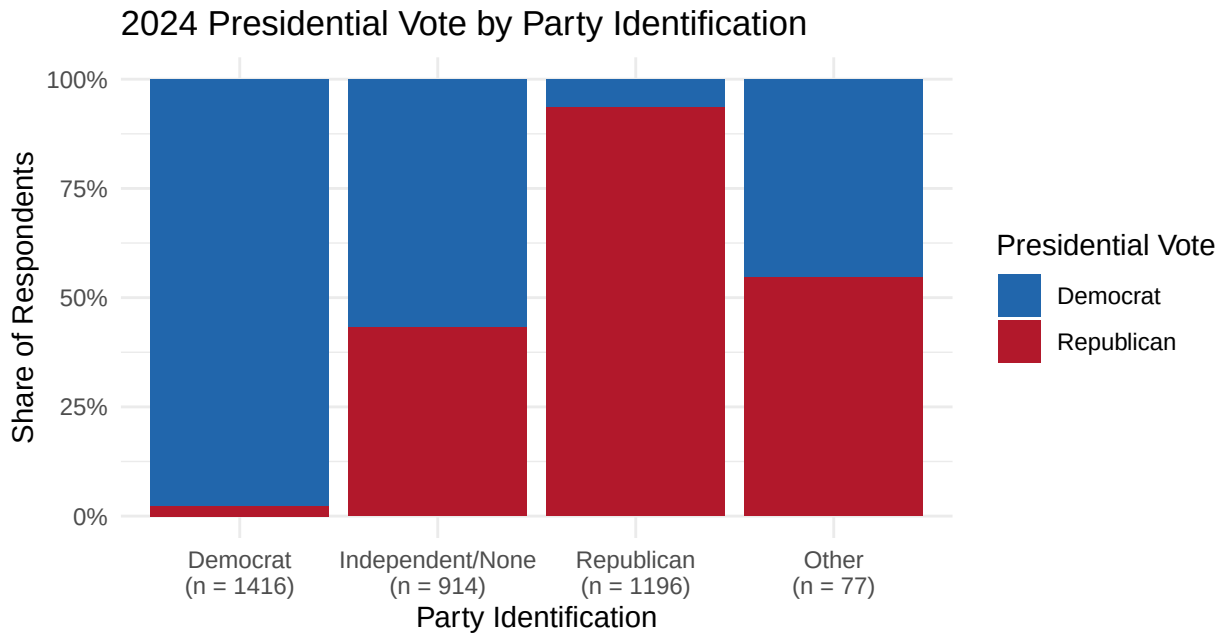
As you can see from the table above, there are rougly ~450 missing values. We chose to drop those values because it would skew the accuracy of the data if we were to put the average or median of the other values in its place. We wanted to keep the data as accurate as possible.

4 Visualization

All plots below show the share of respondents who voted Democratic (blue) and Republican (red) within each group. The number of respondents in each group is printed under its label, because the percentages of small groups are less precise. The helper functions in `01_funct_Plots.R` produce the plots and tests, and the full set of plots, tables and tests is in `31_DataVisualization.Rmd`. The table below gives the chi-square test of independence between each categorical characteristic and the presidential vote.

Variable	ChiSquare	df	p_value
social_class	63.7	3	<0.001
race	178.1	5	<0.001
gender	60.6	3	<0.001
education_level	163.4	3	<0.001
party_id	2189.0	3	<0.001

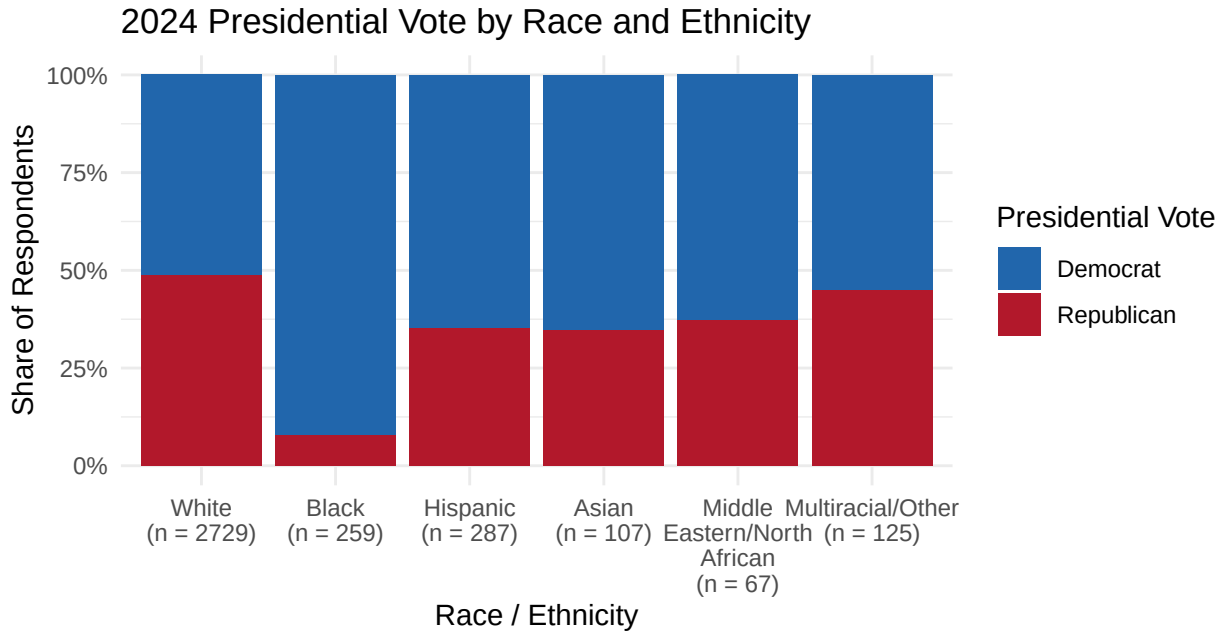
4.1 Party identification



Party identification is very strongly related to vote choice. Almost all Democratic identifiers (97.7%) and Republican identifiers (93.5%) voted for their own party's candidate, and Independent/None respondents split 56.8% Democratic and 43.2% Republican. Only 111 respondents voted against

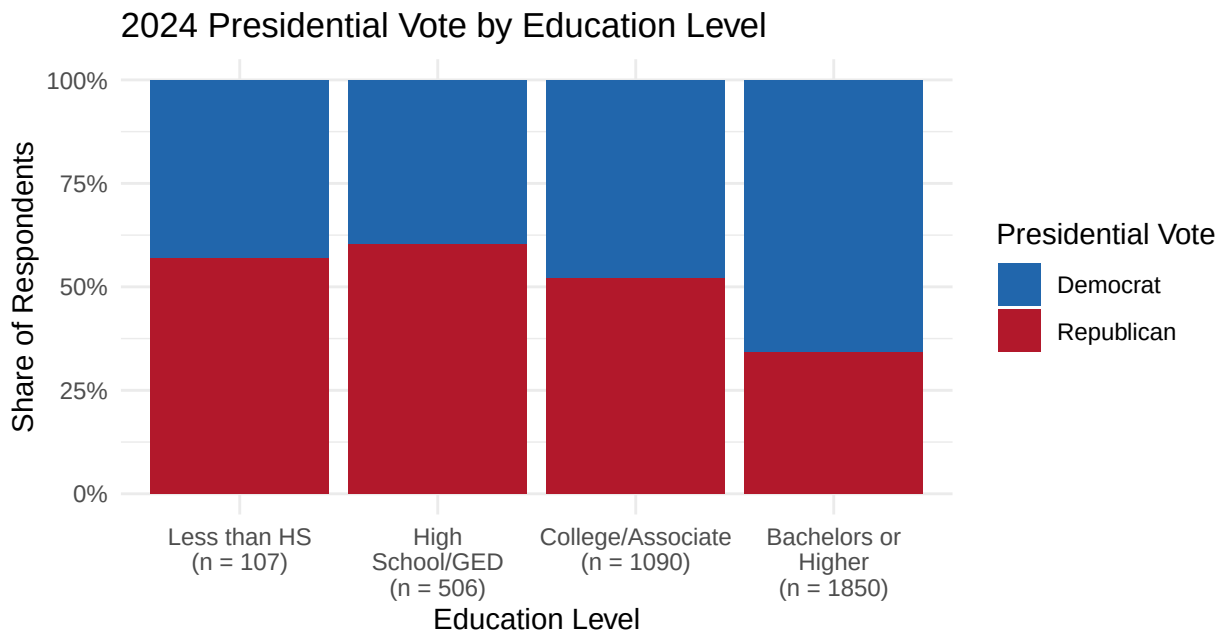
their party. This relationship is so strong that it suggests treating party identification separately from the demographic characteristics in the models.

4.2 Race and ethnicity



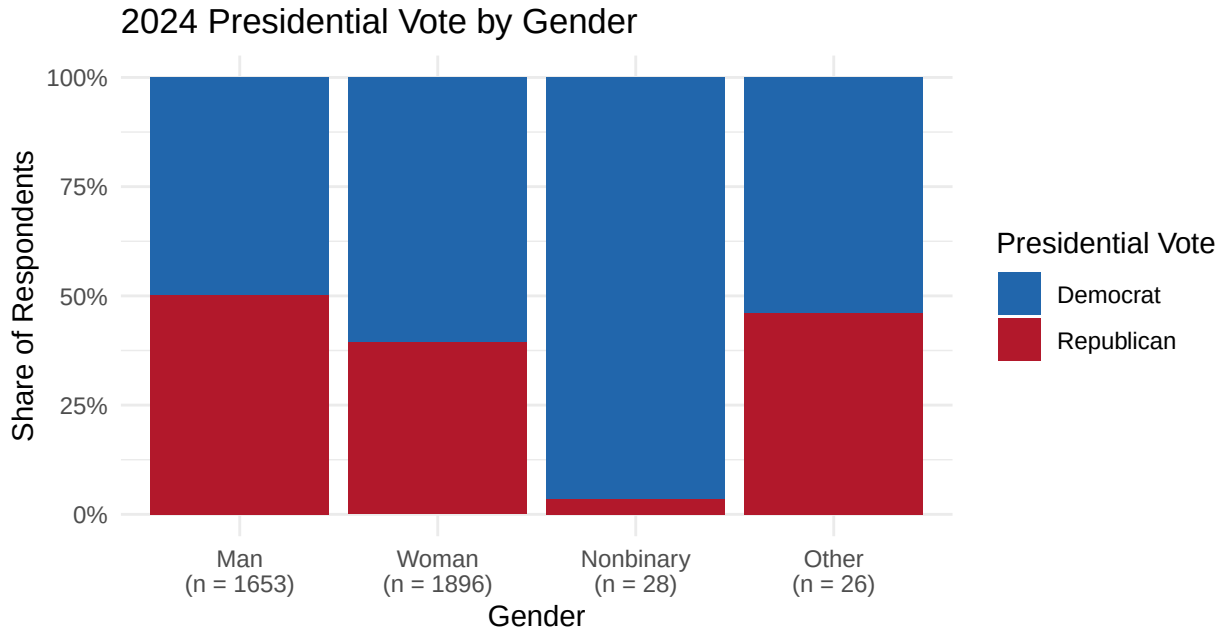
Black respondents were overwhelmingly Democratic (92.3%), Hispanic, Asian and Middle Eastern/North African respondents leaned Democratic (64.8%, 65.4% and 62.7%), and White respondents were nearly evenly split (51.2% Democratic). The differences across groups are large and statistically significant. The Asian, Middle Eastern/North African and Multiracial/Other groups are small (107, 67 and 125 respondents), so their percentages are less certain.

4.3 Education



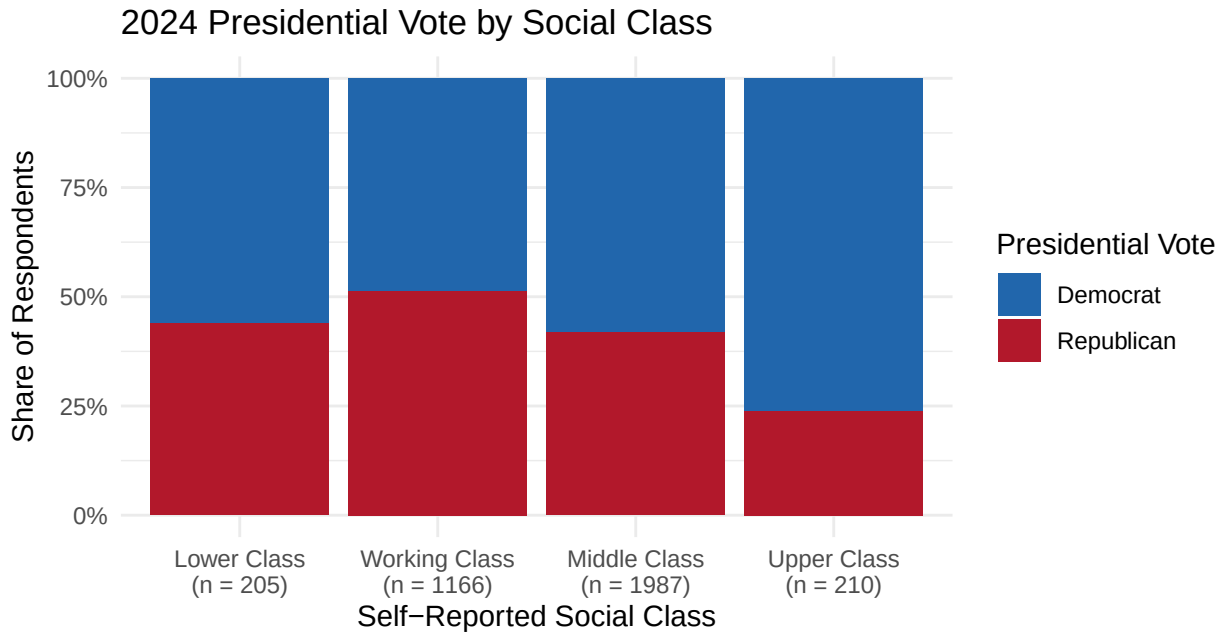
Respondents with a bachelor's degree or higher were the only group with a clear Democratic majority (65.8%); the Democratic share was between 39.7% and 47.9% in every other group. The pattern is not a smooth trend, because the lowest Democratic share is in the high school/GED group and the less-than-high-school group is small (107 respondents). The main contrast is between college graduates and everyone else.

4.4 Gender



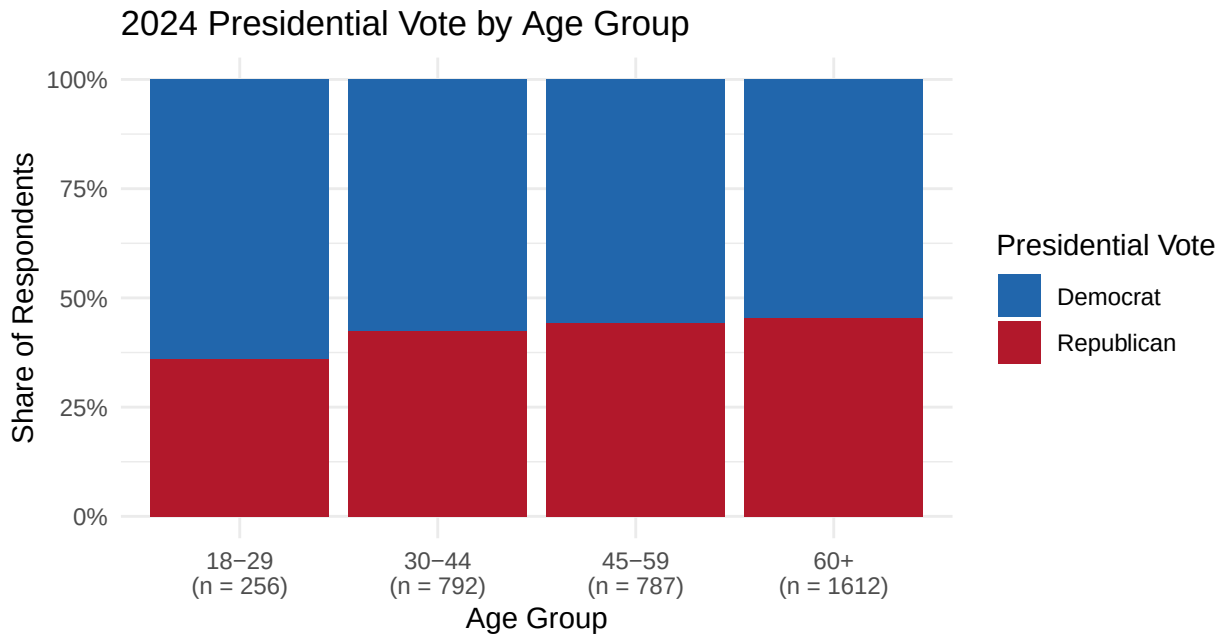
Women were more likely to vote Democratic than men (60.7% compared with 49.8%), and this gap is statistically significant even when the test is limited to men and women. The Nonbinary (28 respondents) and Other (26 respondents) groups are too small to interpret on their own, and "Other" includes every response that was not coded as man, woman or nonbinary.

4.5 Social class



The relationship between social class and vote choice is not a steady trend. The Republican share rises from the lower class (43.9%) to the working class (51.3%), which is the only group with a Republican majority, and then falls for the middle class (41.8%) and the upper class (23.8%). Social class is self-reported, and it is related to education, so it is unclear how much of this pattern remains once education and the other characteristics are taken into account.

4.6 Age



Age has the weakest relationship of the characteristics we examined. Republican voters were slightly older on average than Democratic voters (mean age 56.1 compared with 54.9 years). A Welch two-sample t-test found this difference of about 1.3 years to be statistically significant, $t(3290.9) = -2.21$, $p = .027$, and the age distributions overlap heavily. Grouping age shows that the Democratic share is highest among respondents aged 18 to 29 (64.1%) and falls with each older group (57.6%, 55.8% and 54.6%), $\chi^2(3) \approx 8.8$, $p = .032$.

4.7 What the visualizations suggest and planned further visualizations

Together, the plots suggest that vote choice is closely tied to party identification, strongly related to race, education and gender, and only weakly related to age. They also show that the characteristics are probably related to each other (for example, education and social class), so it is not clear from one-variable plots which differences remain when the other characteristics are held constant. This is the reason for the multivariable models in the next section. For the final report we plan to add the following visualizations.

- A heat map or faceted bar chart of the Democratic share by race and education, and by gender and age group, to look for combinations of characteristics that matter more together than separately.
- A coefficient (odds ratio) plot with confidence intervals for the logistic regression models, with and without party identification.
- Plots of the predicted probability of a Democratic vote across age, education and race for typical respondents.
- An ROC curve and a confusion matrix chart for the cross-validated predictions.
- A comparison of the presidential vote and the House vote, to look at respondents who voted for different parties in the two races.

5 Analysis

5.1 Models

The main model will be a logistic regression of the presidential vote (Democrat = 1, Republican = 0) on the voter characteristics. We will start with a demographic model that uses age, gender, race and ethnicity, education and social class (Model 1), and then add party identification (Model 2). Likelihood ratio tests and AIC will be used to compare the models, and odds ratios with 95% confidence intervals will be used to interpret them. We will also test interactions that the plots suggest, such as race by education and gender by age group, and compare a simpler model with a more flexible classifier such as a classification tree or a random forest to see whether non-linear patterns improve prediction.

5.2 Evaluation

To measure how well the models predict vote choice, we will use 10-fold cross-validation (or a training and test split) and report accuracy, a confusion matrix and the area under the ROC curve. The baseline for comparison is the accuracy of always predicting the larger group, which is 55.9% (Democratic). The comparison between Model 1 and Model 2 will show how much of vote choice demographics alone can explain.

5.3 Robustness and data issues

- **Missing data.** A complete-case analysis would use 3,355 respondents for the demographic model. We will compare it with multiple imputation to see whether leaving out respondents with missing values changes the results.
- **Small groups.** The Nonbinary and Other gender groups and some race groups have few respondents, so we will check whether combining them or dropping them changes the conclusions.
- **Survey weights.** The results so far are unweighted. The ANES provides survey weights, and we plan to repeat the main models with the weights as a sensitivity check.
- **Limitations.** The data are self-reported, they cover only respondents who reported voting for a Democratic or Republican candidate, and an observational survey cannot show that any characteristic causes a vote choice.

5.4 Expected outcomes

We expect party identification to be by far the strongest predictor of vote. Among the demographic characteristics, we expect race (especially Black respondents), education (a bachelor's degree or higher) and gender to remain significant after adjusting for the other variables, and we expect the age effect to be small. We expect the non-linear social class pattern to become weaker once education is included, which would suggest that social class partly reflects education. If the demographic model predicts vote only slightly better than the 55.9% baseline while the model with party identification predicts it very well, this would show that these demographic characteristics alone are limited predictors of individual vote choice.

5.5 Contributions

Eva: Wrote the Readme files and did the data cleaning and some of the processing for the dataset. Created 11_DataCleaning and 12_RData and 00_requirements. Found the dataset, created the repository, proposed the idea for the project. Added some explanation for dropped/missing values per TA instruction. Dennis: I contributed to the data analysis in 13_Analysis.Rmd. I also restructured the project workflow by separating the data processing and data visualization steps into their own files (21_DataProcessing.Rmd and 31_DataVisualization.Rmd), which made the pipeline easier to follow and maintain. In addition, I made several improvements to the data cleaning file (11_DataCleaning.Rmd) and fixed issues in the final report (FinalReport.Rmd) Kenneth:

5.6 Github Link : https://github.com/eevs23/STAT107Final